

# Infinite Tokens, Finite Atoms

Scaling Gen AI demand within the constraints of Chips, Watts and Dollars

## Technology

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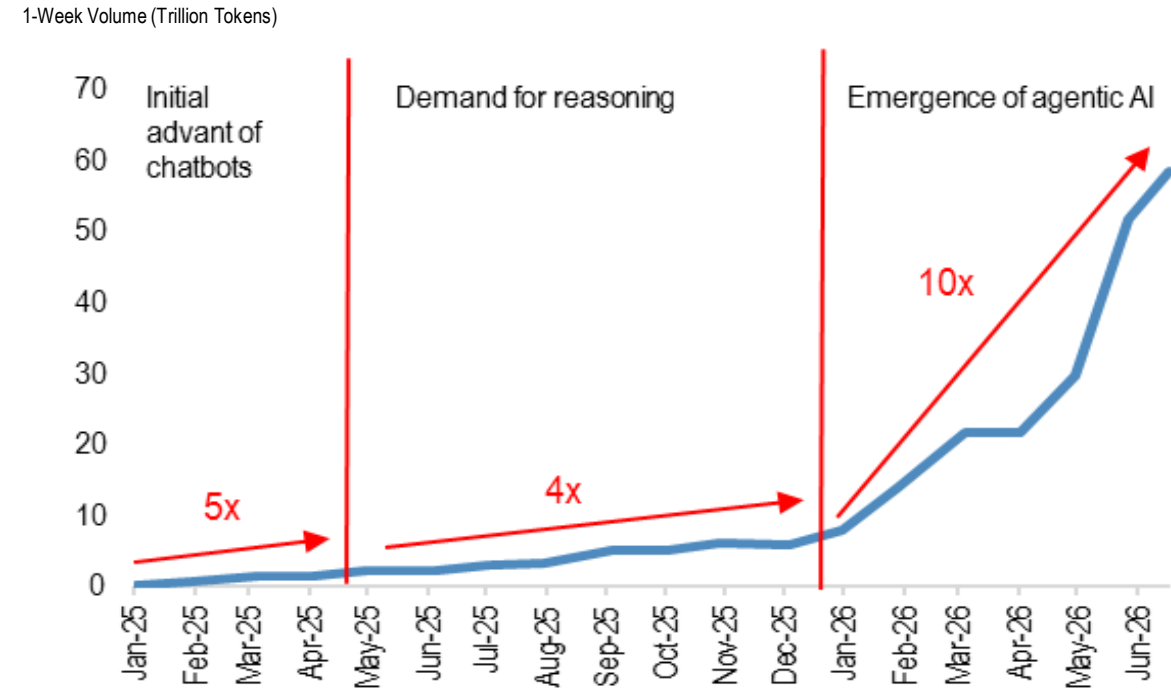
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# Token demand growth continues to accelerate

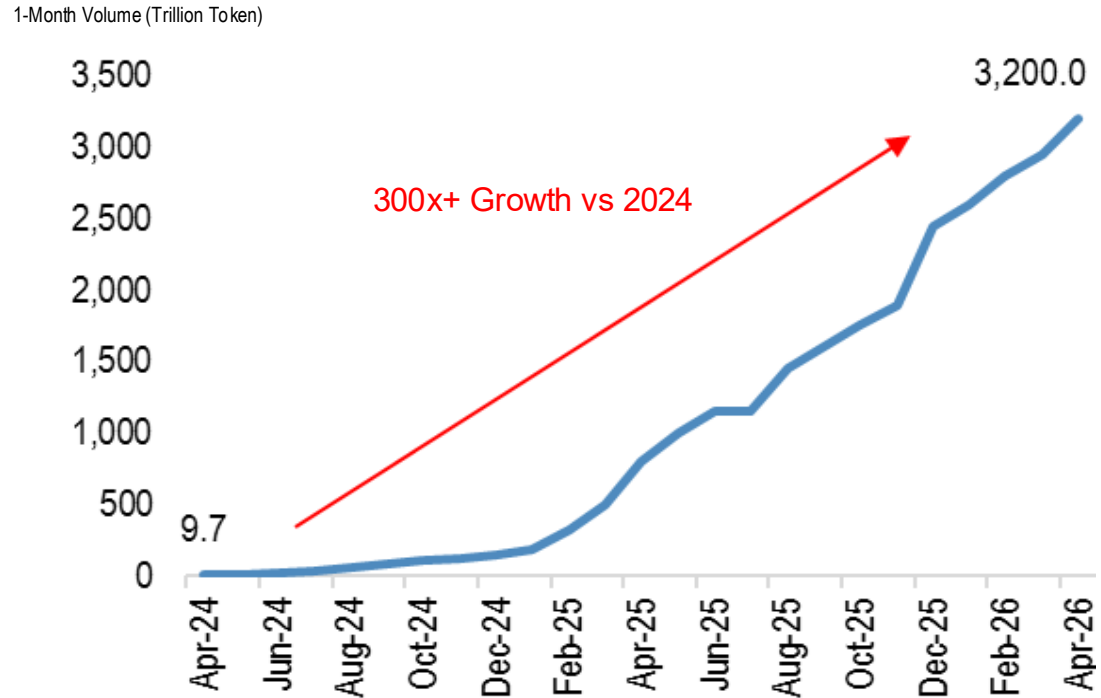
- Token growth accelerating from Chatbots era to Reasoning Models to Agentic AI
- All indicators show acceleration in growth rates, rather than deceleration – even with large numbers in play

Stages of token demand acceleration



Source: OpenRouter, J.P. Morgan estimates.  
Note: OpenRouter is a directional indicator, rather than total volume given it doesn't capture large parts of proprietary AI Model token consumption

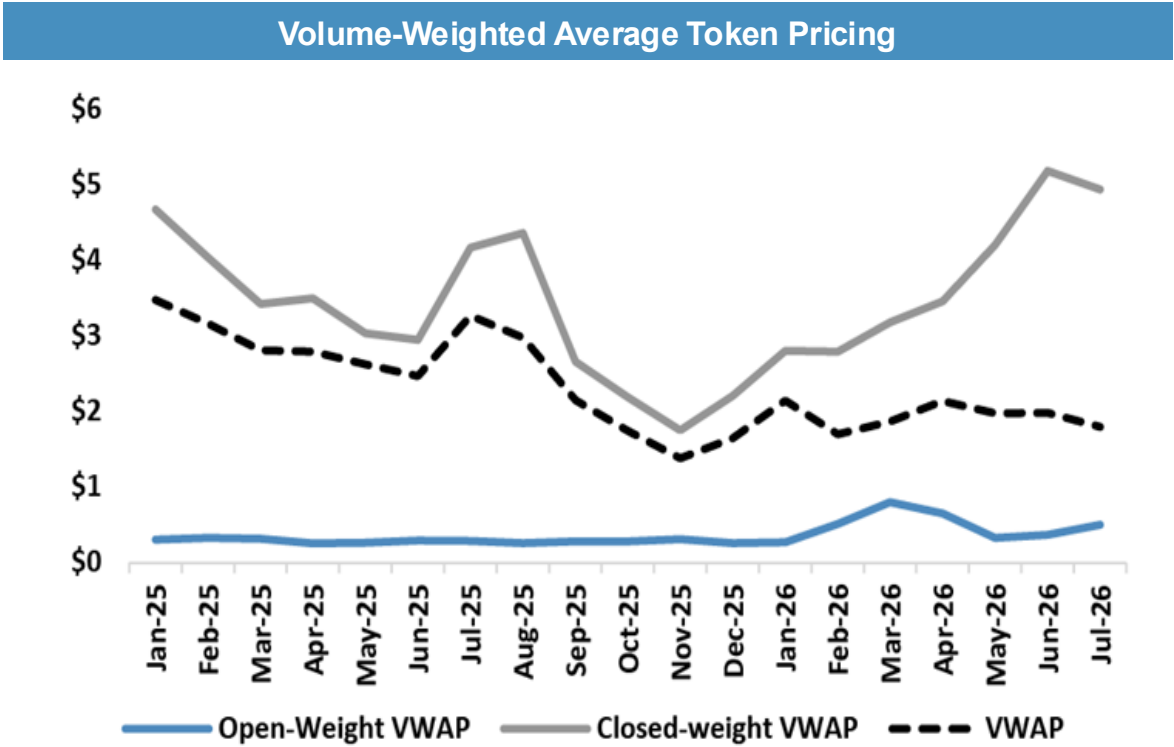
Total monthly tokens processed across Google surface



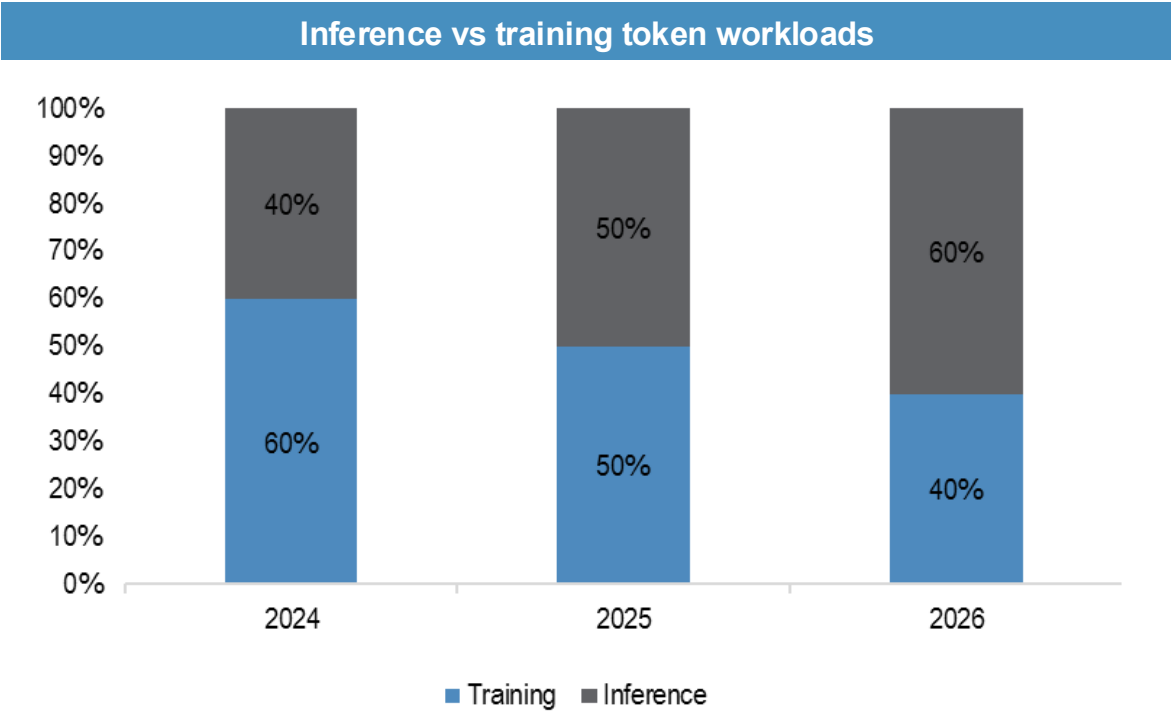
Source: Google; J.P. Morgan estimates.  
Note: Google Surface including AI Overviews, AI Mode, Agent Platform, AI Studio, Gemini App, Workspace, Antigravity, Other

# Open Source taking more share of tokens, proprietary models still dominate revenues

- Open-Source Tokens taking more market share, driving blended token pricing down and increasing overall GenAI adoption.
- Proprietary Frontier models still growing fast, in a compute-constrained world
- Mix leaning more towards Inference – led by Agentic, but RSI could change this again



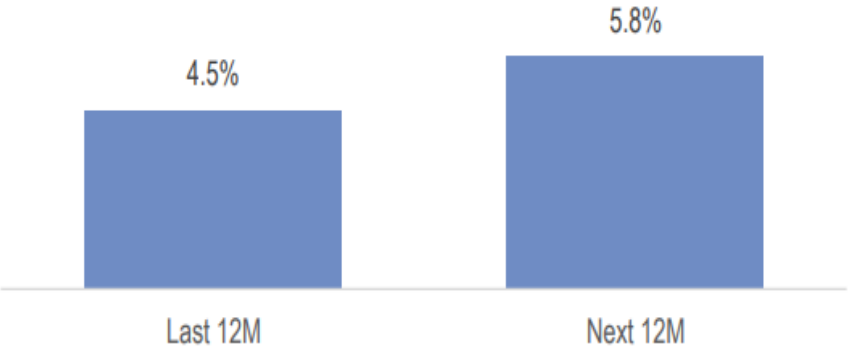
Source: OpenRouter, J.P. Morgan Estimates



Source: AMD.

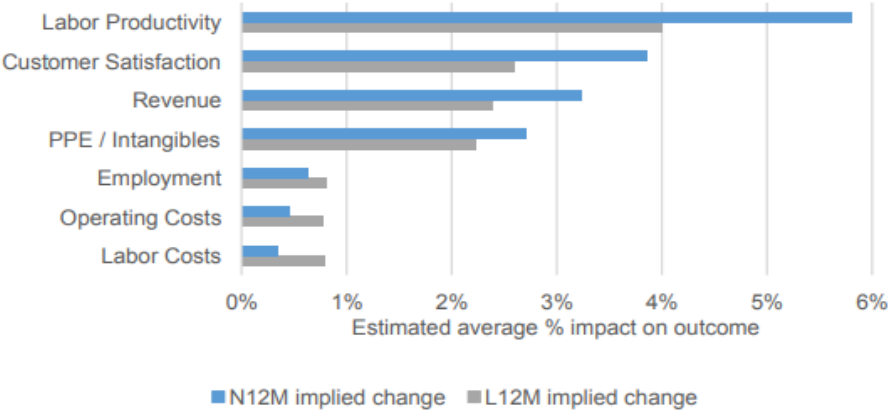
Early days, but strong adoption trends in the enterprise, across different verticals

JPM AI survey – Avg AI spending as% of expenses



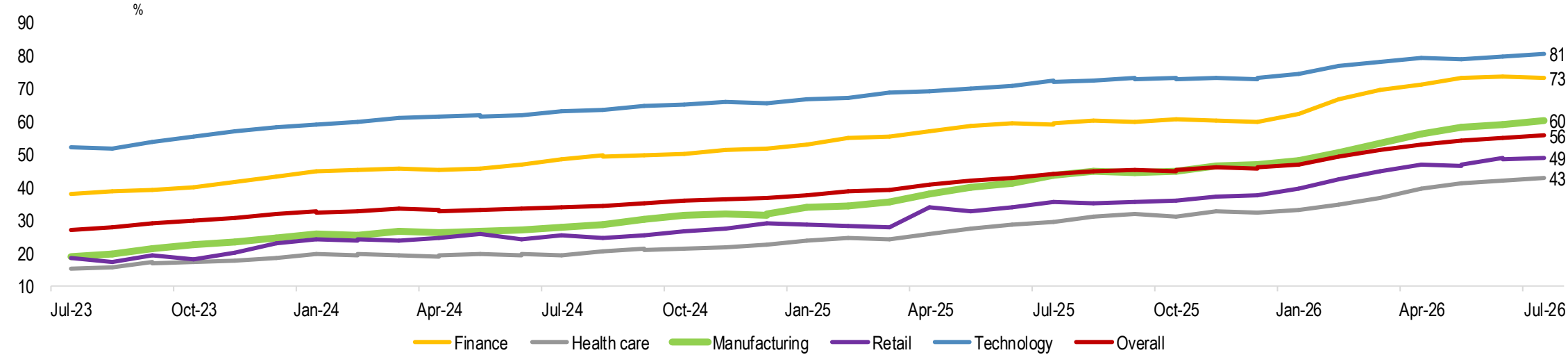
Source: J.P. Morgan.

JPM AI survey – Where can AI make a difference?



Source: J.P. Morgan.

Ramp AI Index: Share of U.S. businesses with paid subscriptions to AI models, platforms, and tools (by sectors)



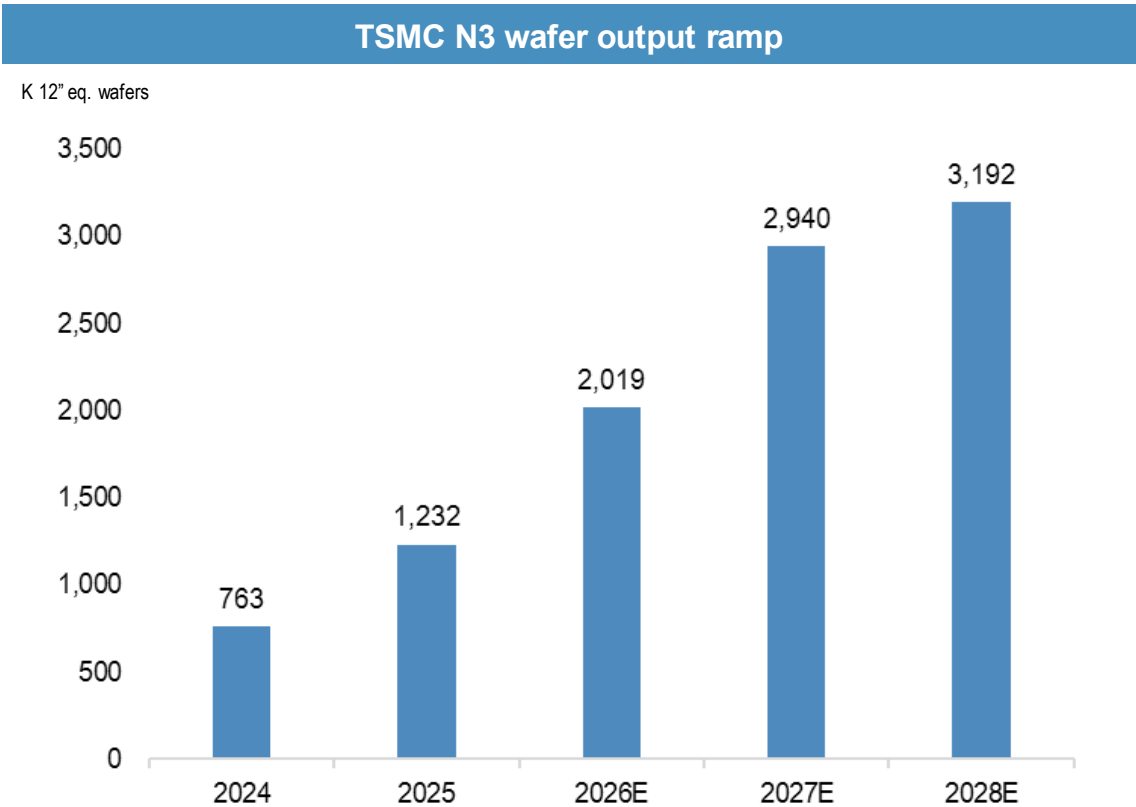
Source: Ramp AI.

1. Compute:

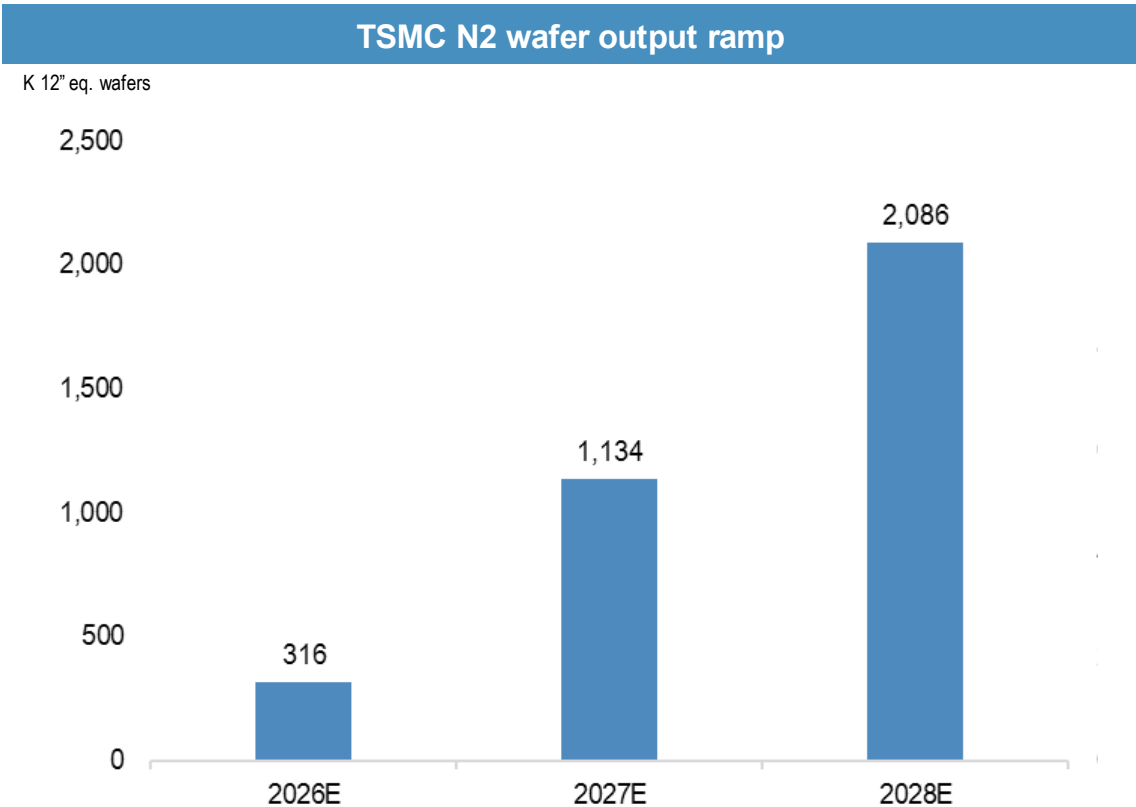
Near-term constraint for AI infra deployment

# Compute– Leading edge wafers are the primary bottleneck now

- Leading edge wafer capacity – N3 and N2 – remain the primary bottleneck for compute demand currently.
- Upside from Agentic AI CPU and AI ASIC demand.
- TSMC capex accelerating, but unlikely to reach supply-demand balance before 2028.



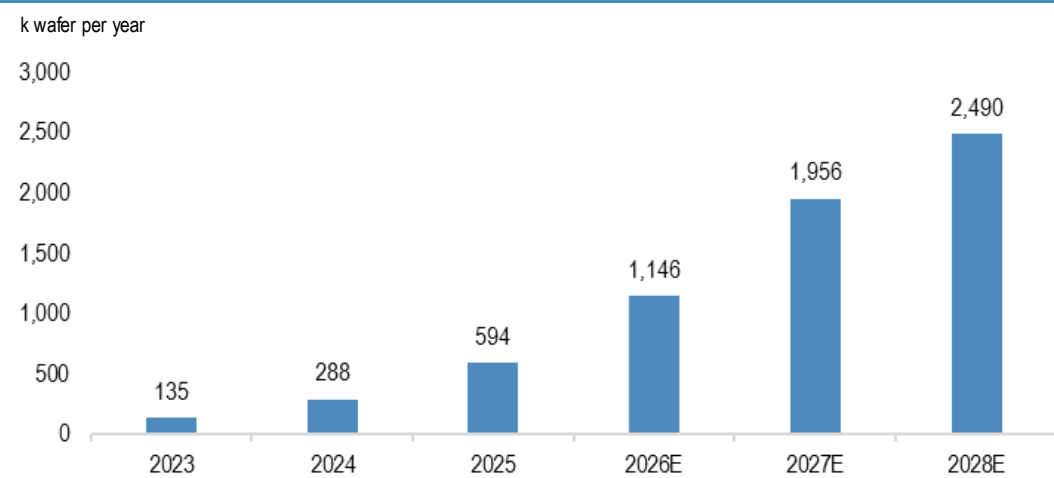
Source: J.P. Morgan estimates.



Source: J.P. Morgan estimates.

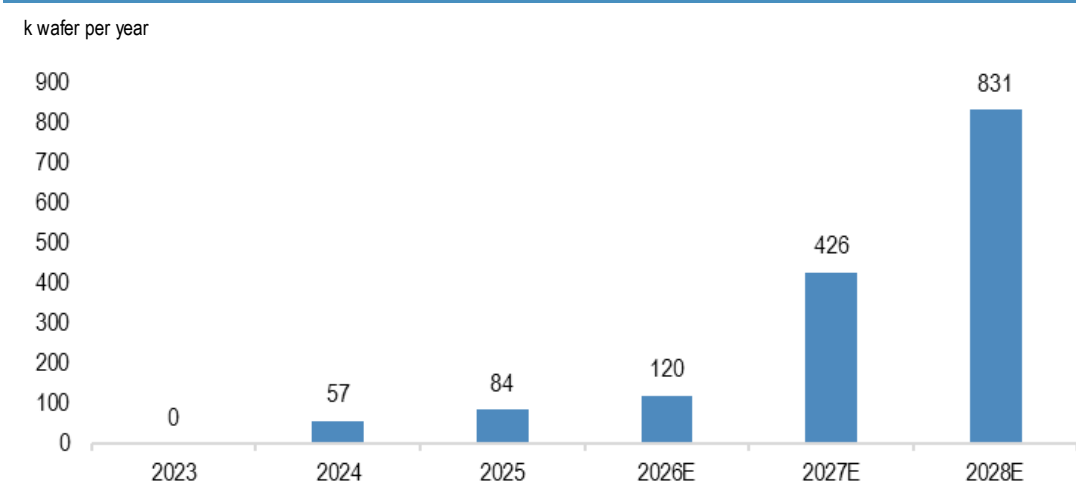
Compute – CoWoS and SoIC capacity the next bottleneck, even with OSAT expansion

TSMC CoWoS capacity



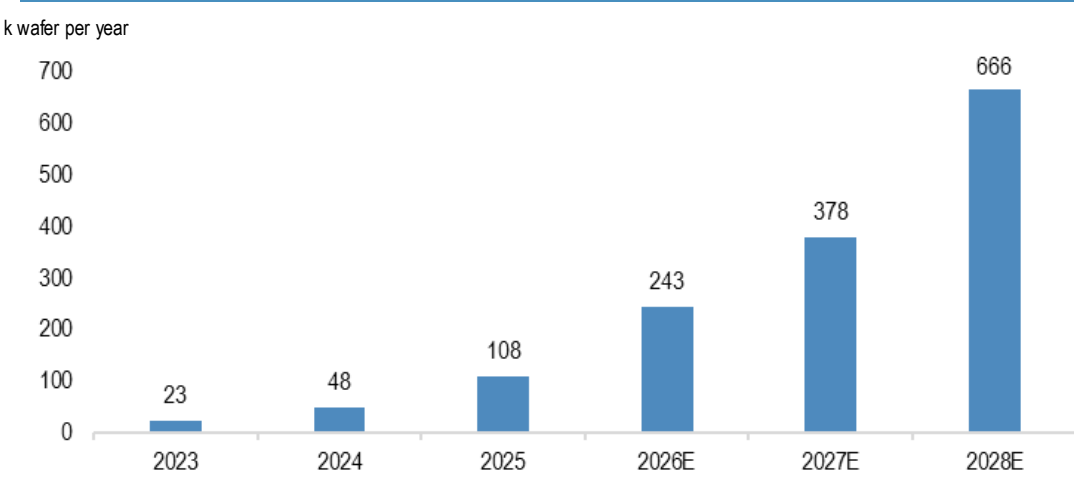
Source: Company data, J.P. Morgan estimates.

Non-TSMC CoWoS capacity



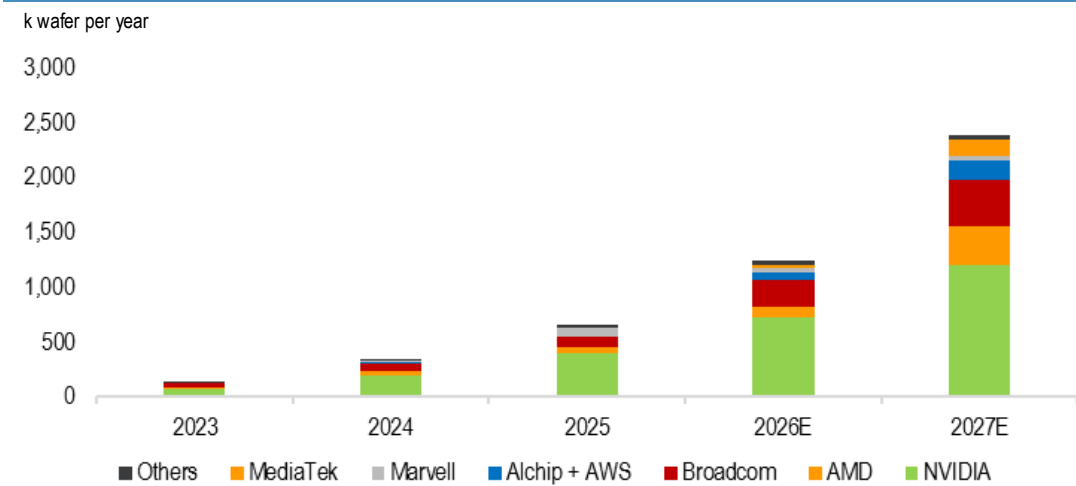
Source: Company data, J.P. Morgan estimates.

TSMC SoIC capacity



Source: Company data, J.P. Morgan estimates.

CoWoS wafer allocation by customers

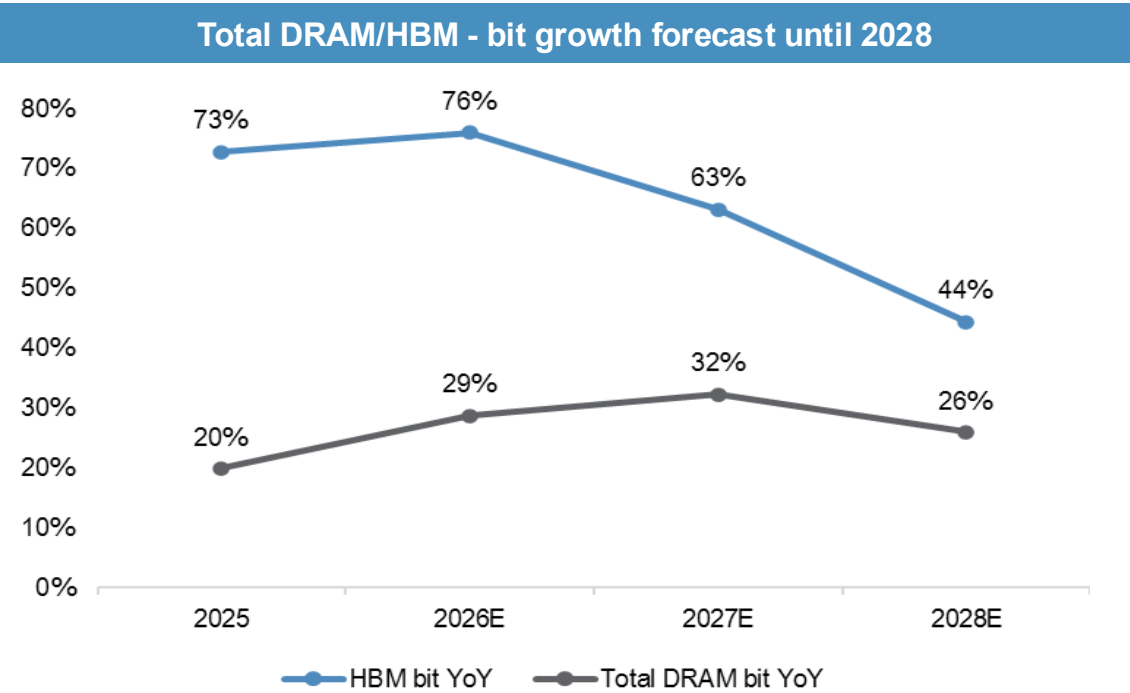


Source: Company data, J.P. Morgan estimates.

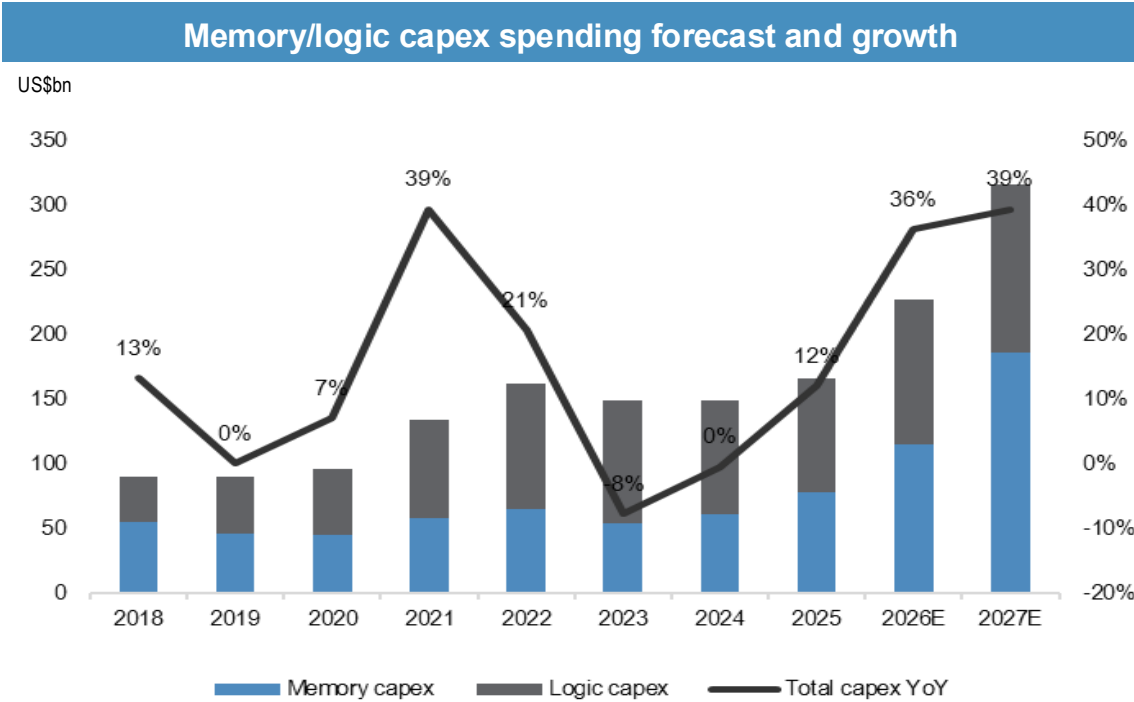
Memory is the next bottleneck, but recent HBM despec + offloading to NAND helps.

- HBM I still in short supply, but recent trends towards HBM de-specing may ease the supply to some extent.
- Agentic AI driving strong demand for commodity DRAM – likely to continue into 2027.
- Offloading of KV cache and demand for cost-efficient storage class to drive upside in NAND

Capex response is starting – but still unlikely to be enough to meet demand before 2028



Source: iSuppli, Gartner®, WSTS, J.P. Morgan estimates



Source: Company data, Gartner® and J.P. Morgan estimates.

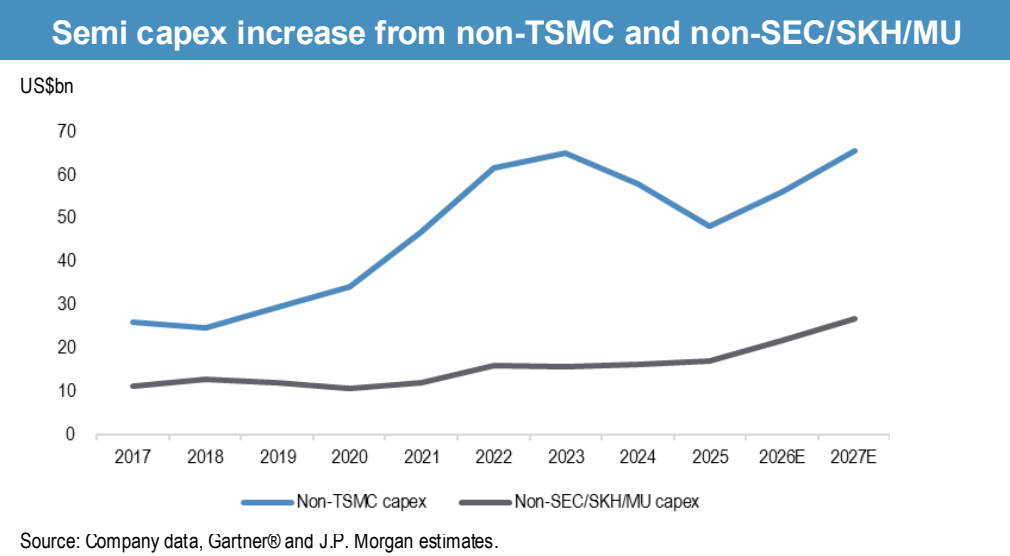
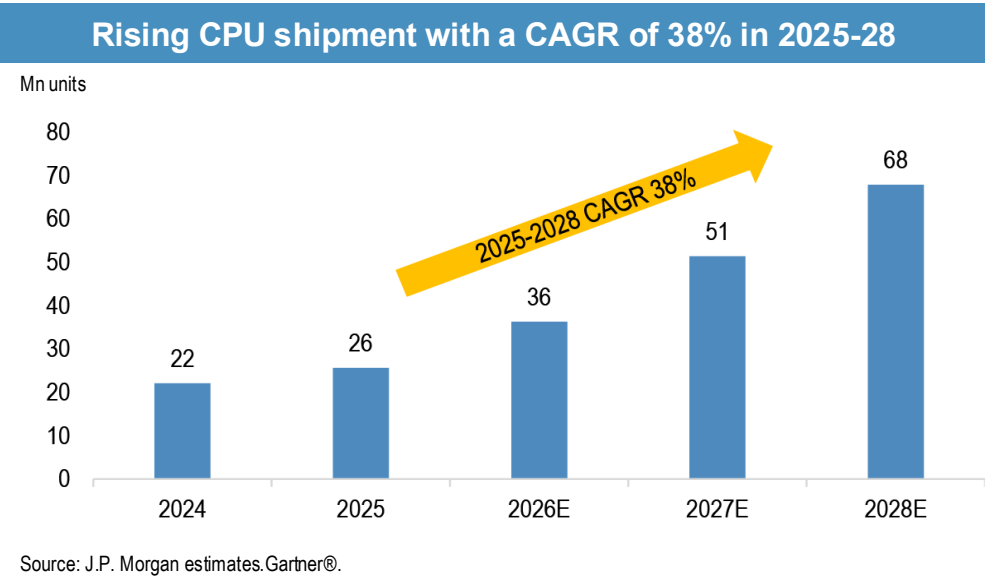
Agentic CPU demand upside and new suppliers joining the fray are key swing factors

■ Key positive influencers in demand

- Agentic AI CPU demand largely unfulfilled in 2026, likely to keep growing in 2027.
- ASIC demand growing much faster in 2027

■ New sources of Supply and Efficiency

- Focus on Efficiency
  - Algorithmic improvement providing 3-4x gains each generation for LLMs
  - Despeccing memory, improve GPU cluster efficiency through Interconnect upgrades.
- New foundry and Memory vendors adding capacity to enter the market.

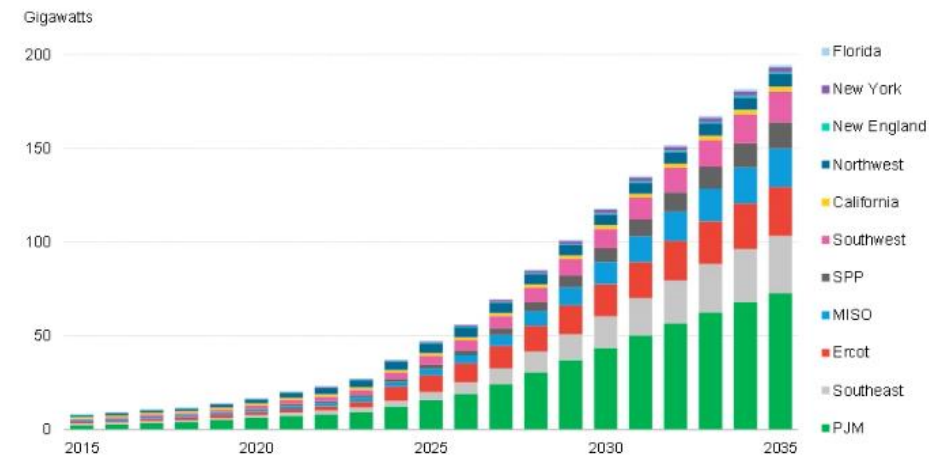


## 2. Datacenter Power:

Strong ramp into 2030 despite many constraints

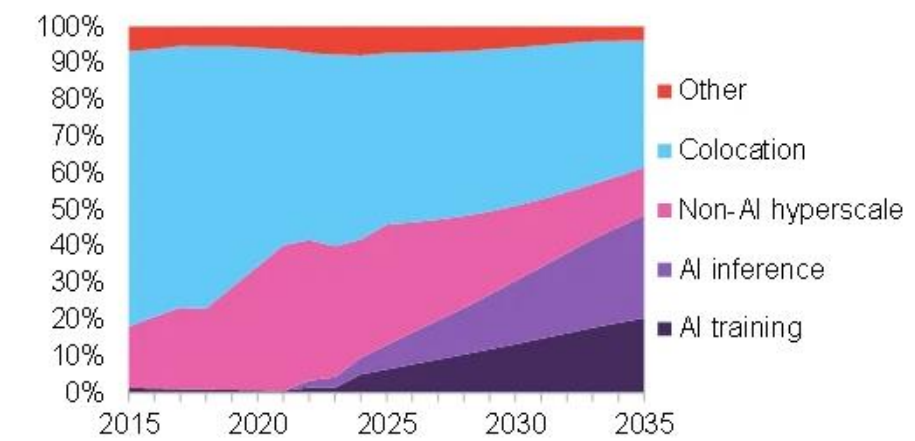
Datacenter Power capacity growing into 2030, forecasts continually revised up

Installed DC capacity in the US – 2x into 2030, 4x into 2035



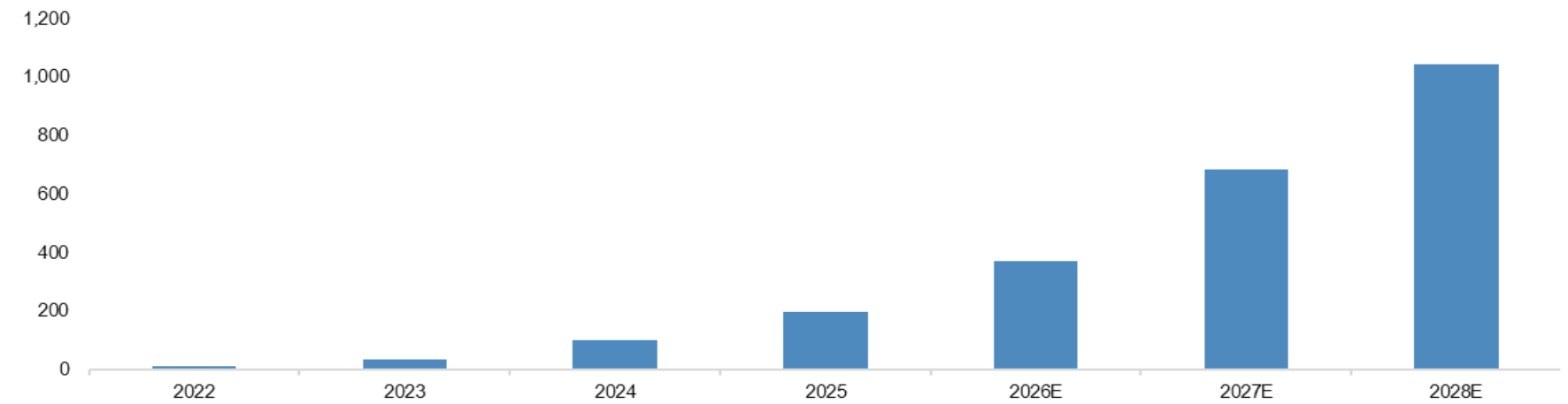
Source: BloombergNEF, DC Byte.

DC workload mix by operator type – AI main growth driver



Source: BloombergNEF, DC Byte.

Total electricity consumption from AI datacenters (TWh)

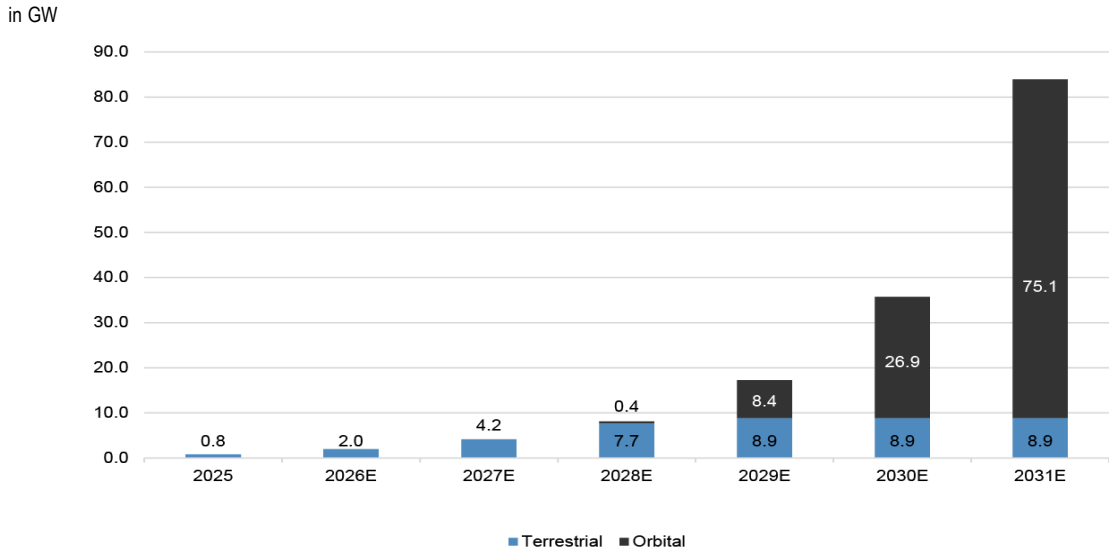


Source: J.P. Morgan estimates.  
Note: assuming the UTR is 65%

Power is not an imminent constraint – but need to watch in 2028-29

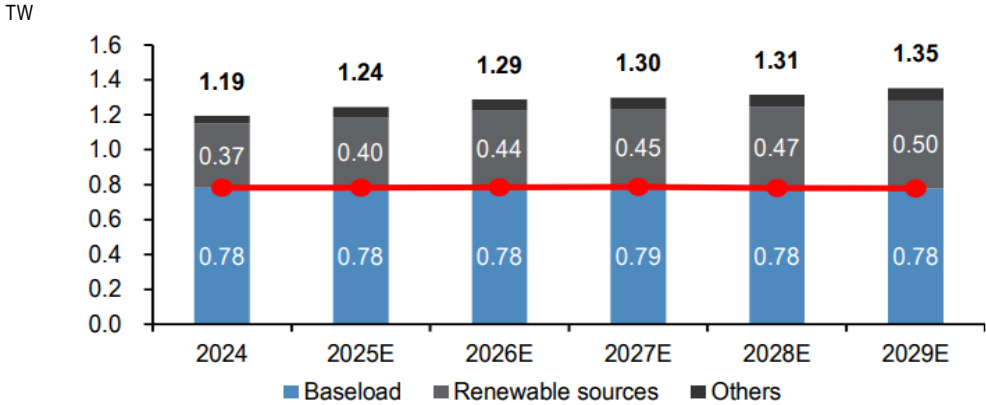
- Grid capacity is maxing out in the US – this is well known and new builds will take time
- Behind the meter solutions are expanding rapidly
- Conversion of other projects (Bitcoin mining, industrial uses) to AI Datacenters
- Datacenters in other locations outside the US, incl. in space.
- **Constraint watch** – sometime in 2028-29 – when the bottleneck shifts from Chips to Power

SpaceX Terrestrial and Orbital Capacity, 2025-2031E



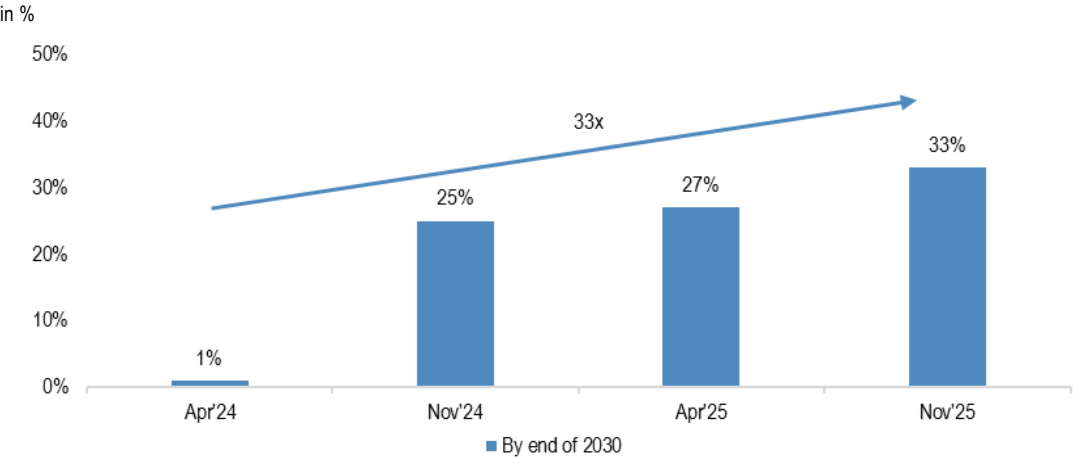
Source: Company data, J.P. Morgan estimates.

Flattish Base-load capacity growth strains grids



Source: EIA, J.P. Morgan.

Expected share of data center with 100% onsite generation



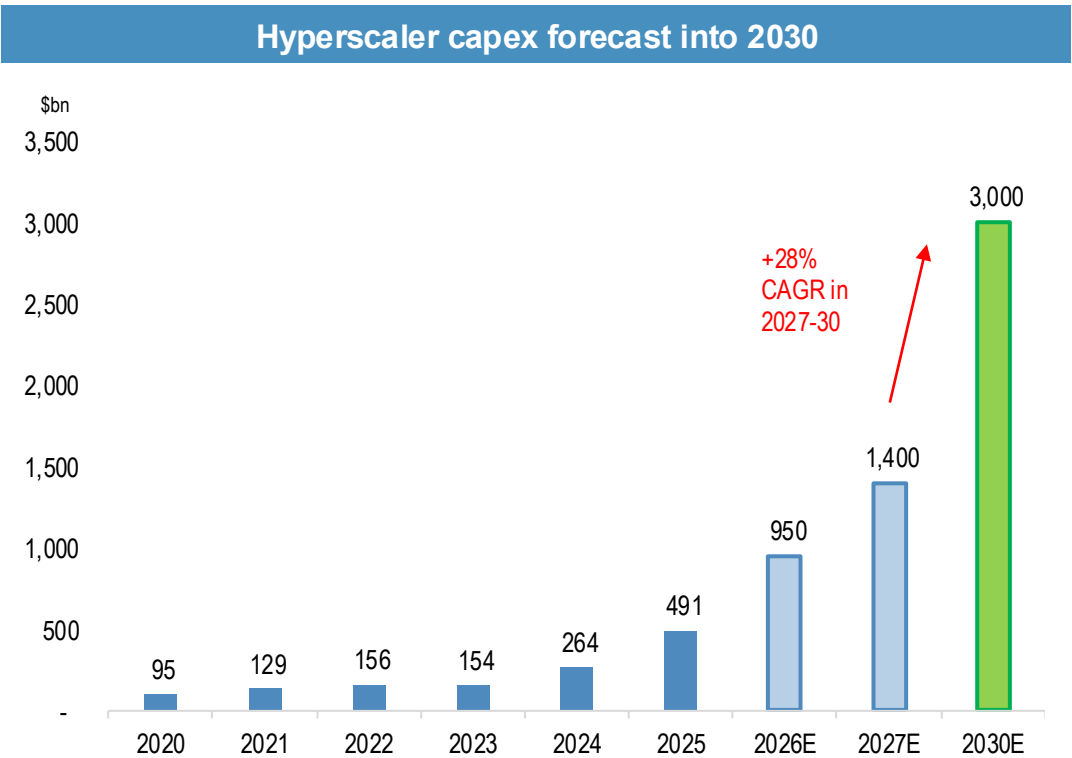
Source: Bloom Energy developer survey (Apr'24–Nov'25); expected share of data centers with 100% onsite generation by YE2030.

### 3. Capital:

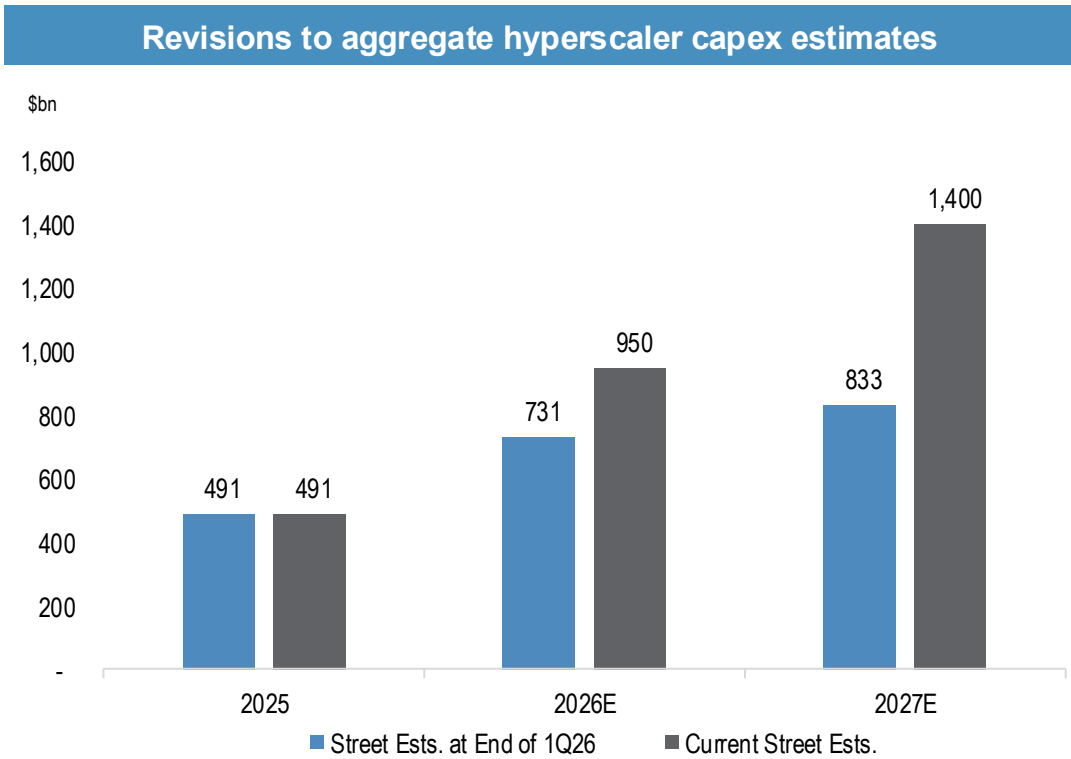
An area of worry currently, but still got headroom

# Hyperscaler capex likely to keep rising into 2028 and beyond

- Hyperscaler capex to more than double in 2026, continued growth into 2028 and beyond.
- Upper reach of hyperscaler capex numbers reach \$3Tn by 2030E,
- Market has been underestimating hyperclear capex systematically in the last few years
  - Leaves more room for upward revisions in future years.

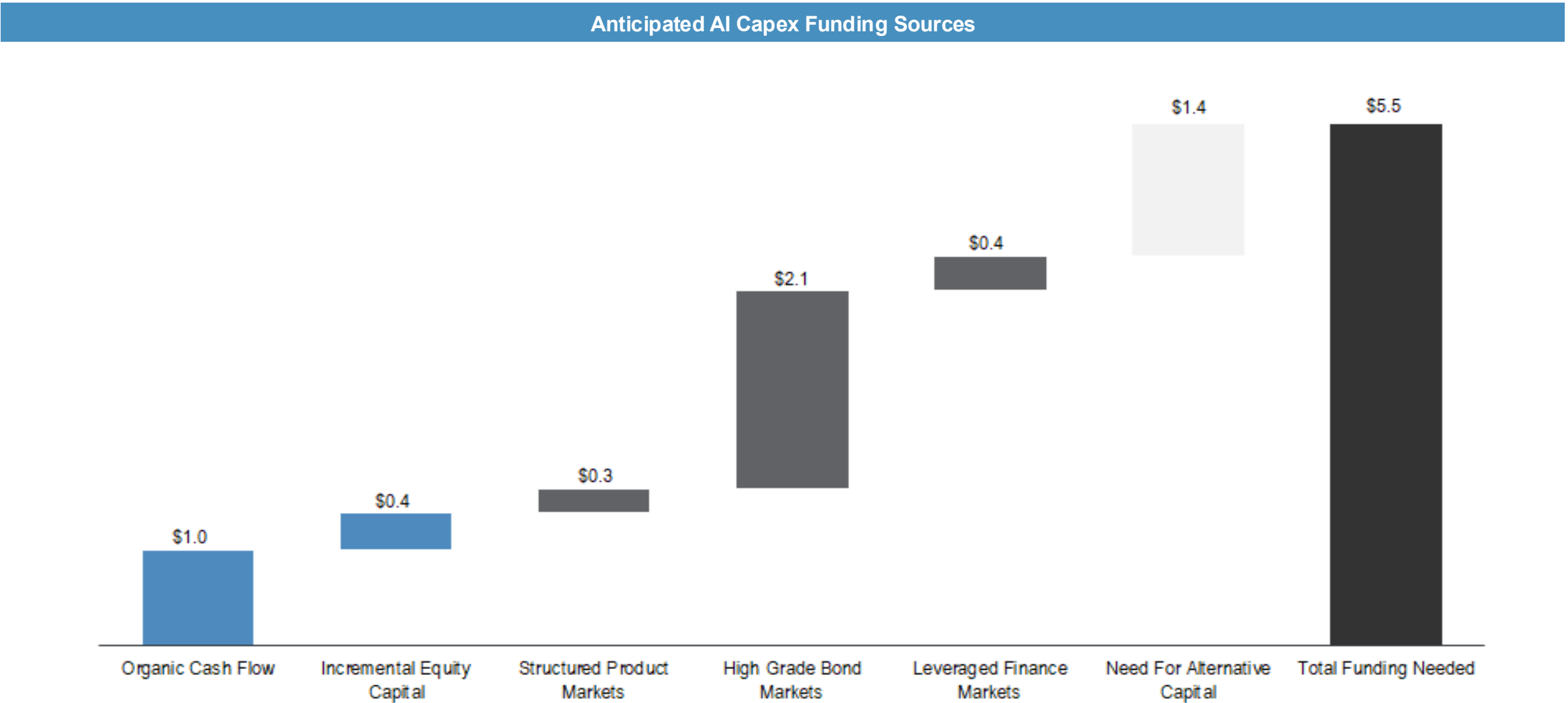


Source: J.P. Morgan Cloud Capex Dataset, Bloomberg Finance L.P.  
Note: Our capex dataset accounts for several cloud/neocloud operators including: Amazon, Google, Microsoft, Meta, Oracle, CoreWeave, Nebius, IREN, and SpaceX; All forward estimates are based on median consensus estimates



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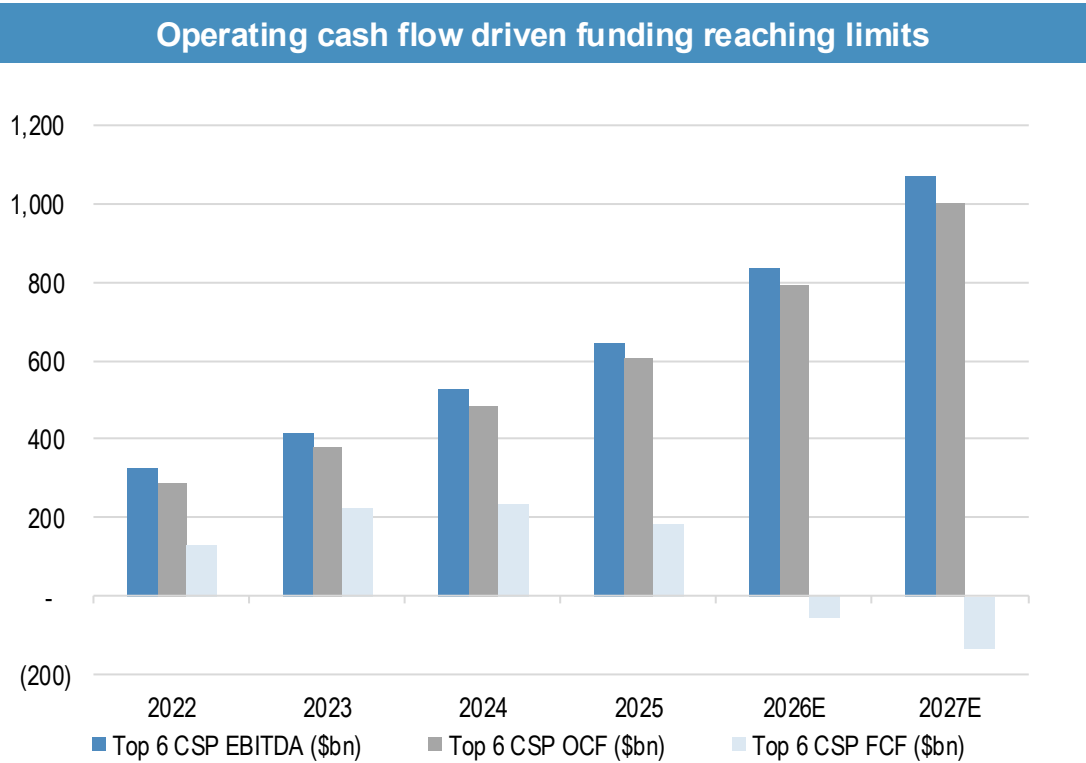
Hyperscaler capex – Sources of funding in the next few years



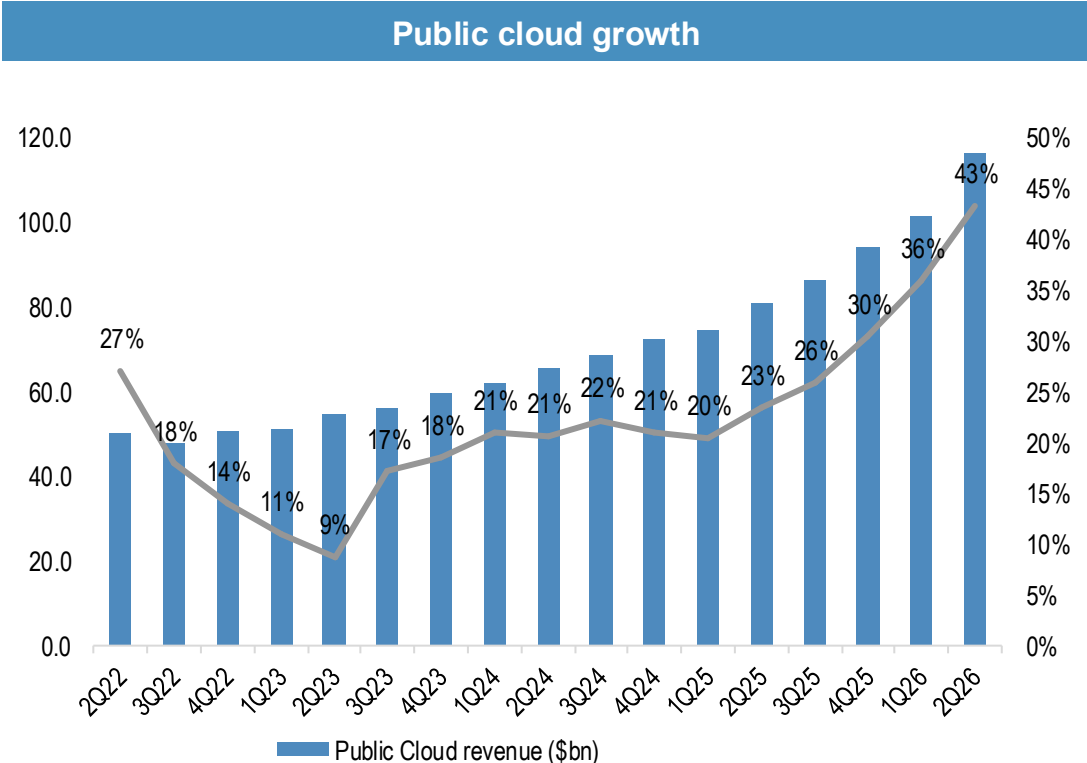
Source: J.P. Morgan.

# Operating cashflow growing, helped by re-accelerating Public Cloud growth

- Operating cash flows financed all of the capex until now, but most hyperscalers are dipping into negative Free Cash Flow from 2027.
- Public Cloud growth is accelerating – a promising sign for accelerating Operating cashflow growth.



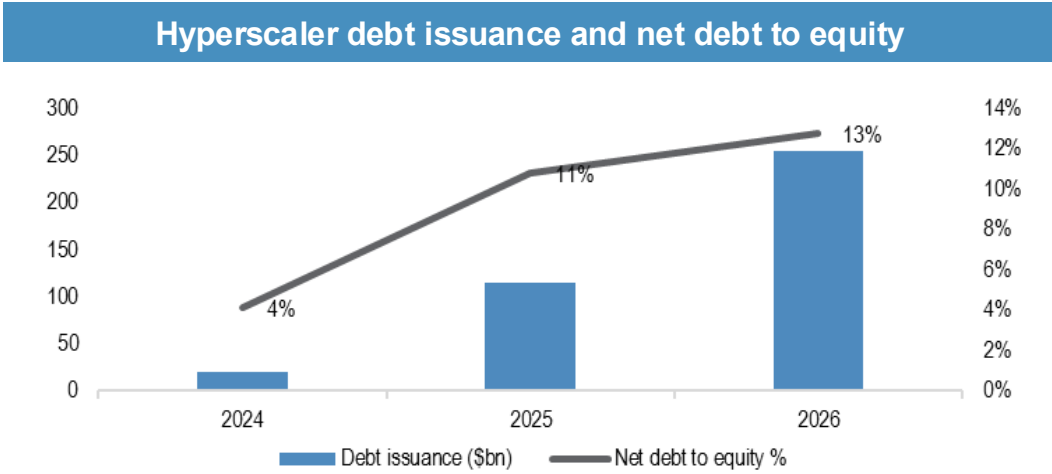
Source: J.P. Morgan estimates, Bloomberg Finance L.P.



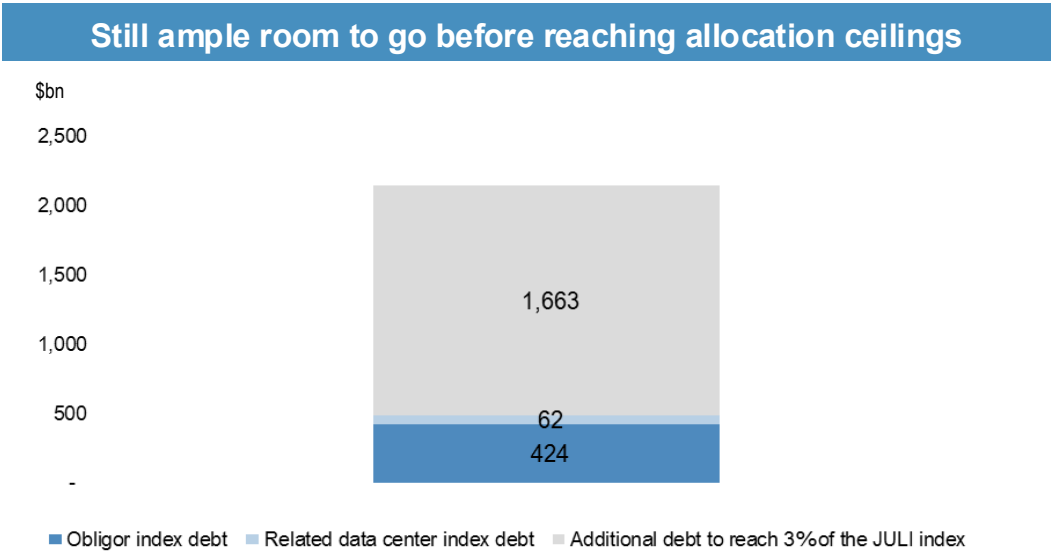
Source: Bloomberg Finance L.P.  
Companies included are Google, Microsoft, Amazon, and Oracle

Debt Funding – Increasing, Concerning, But still got more headroom

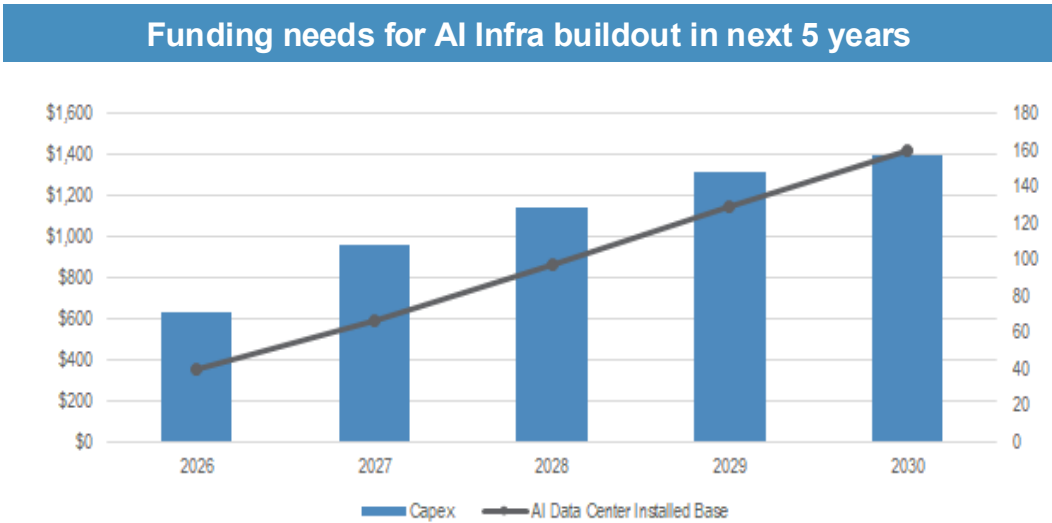
- Debt issuance has picked up meaningfully in 2026.
  - But Net/Debt to Equity is still manageable for hyperscalers.
- Ceiling for IG Debt issuance still quite far
  - Additional \$1.7 trillion of room left before any allocation concerns pop up.
- New financing schemes by GPU vendors enable more access to capital for Neoclouds



Source Bloomberg Finance L.P.  
Note: Top hyperscalers include: Google, Meta, Microsoft, Amazon, ORCL, Coreweave, SpeceX



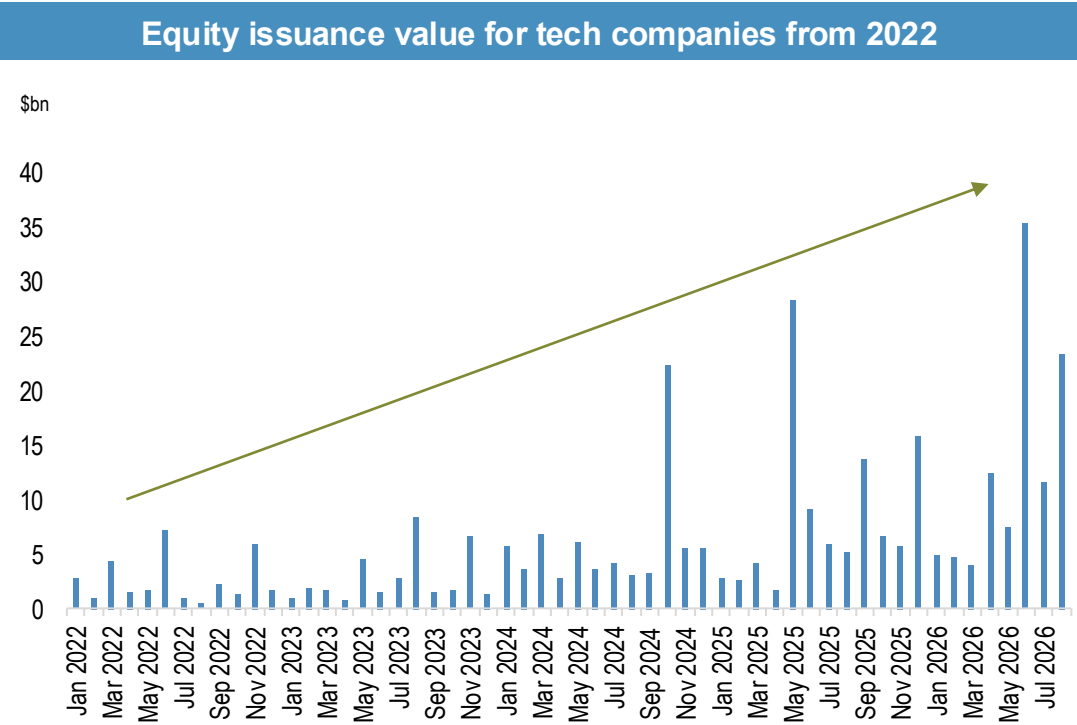
Source: J.P. Morgan, Bloomberg Finance L.P.



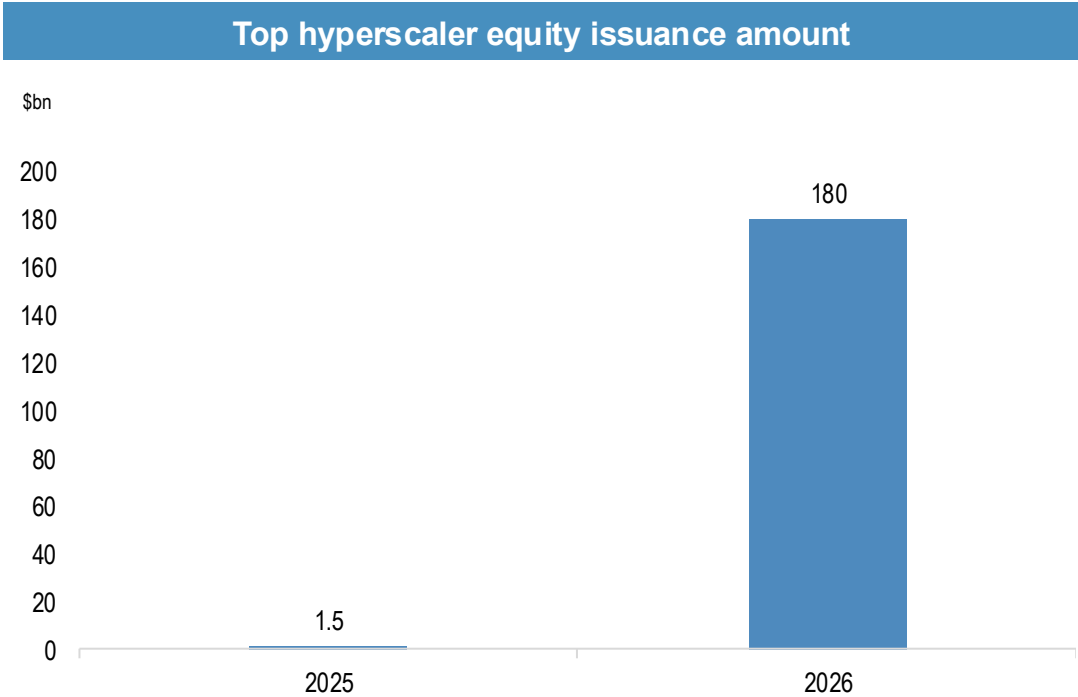
Source: J.P. Morgan.

# Equity issuance is starting, likely could see more momentum

- AI hyperscaler equity issuance surge since 2025
- Top hyperscaler equity issuance topping \$180bn in 2026 YTD.
- New business models coming to the public markets for AI end-Applications creates more room for capital.



Source: Bloomberg Finance L.P. (24 Aug 2026)



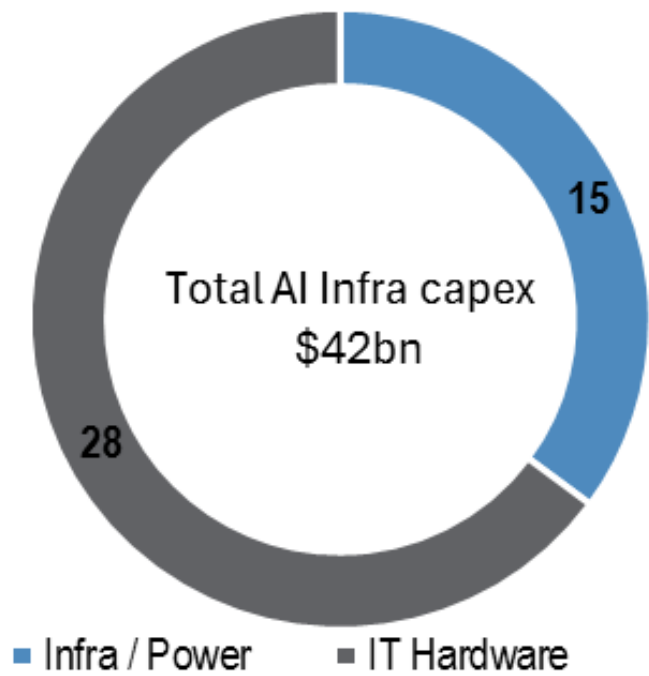
Source: Company reports.  
Note: Top hyperscalers include: Google, Meta, Microsoft, Amazon, ORCL, Coreweave, SpeceX

## The economics of Gen AI Infra buildout

How much does it cost to build 1GW of AI Infra ?

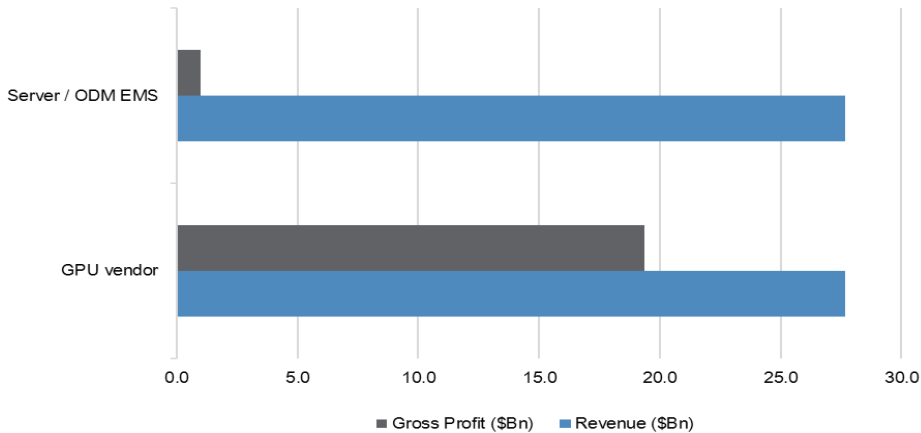
1GW DC capital spend split

US\$bn



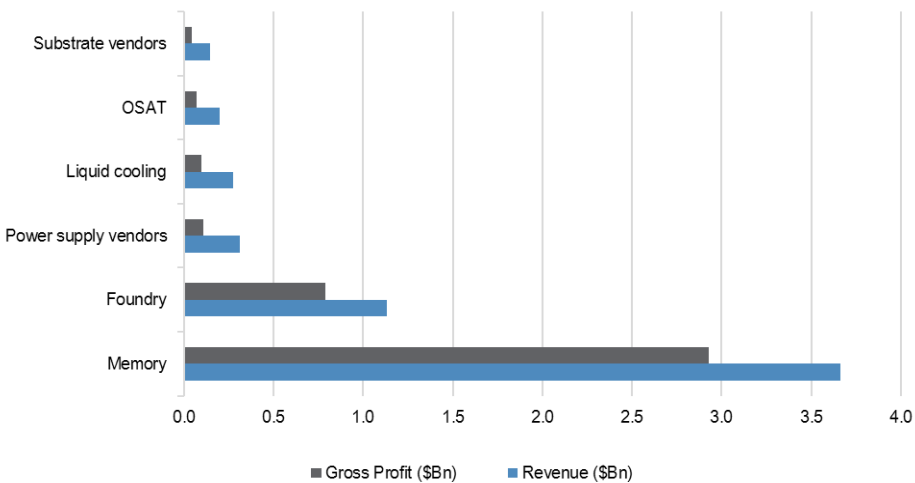
Source: J.P.Morgan estimates.

How much do System/GPU vendors earn from 1GW



Source: J.P.Morgan estimates. Note: Assuming that GPU vendor is recognizing full rack scale revenues.

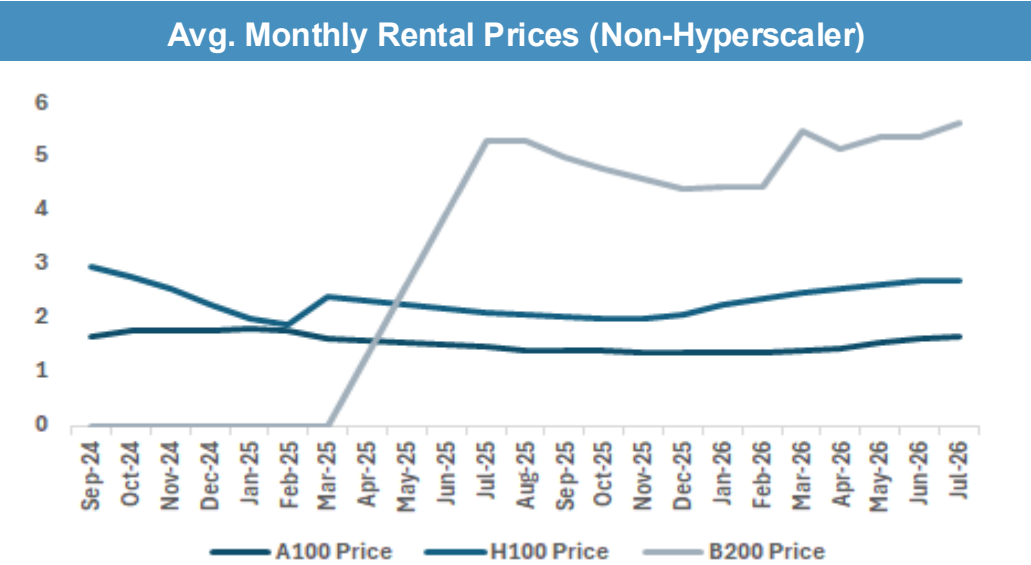
How much do component vendors earn from 1GW



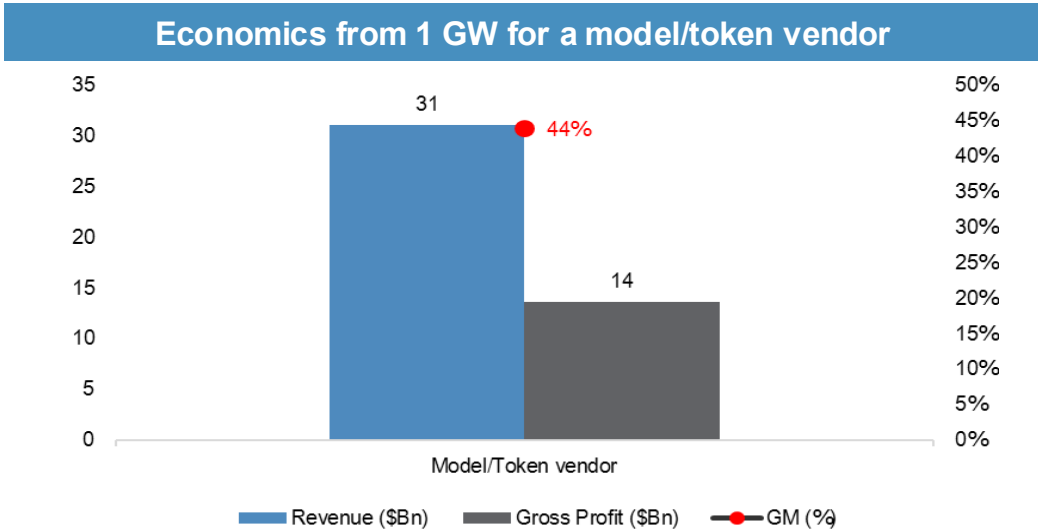
Source: J.P.Morgan estimates.

# What can 1GW of AI Compute generate from GPU rentals and token sales?

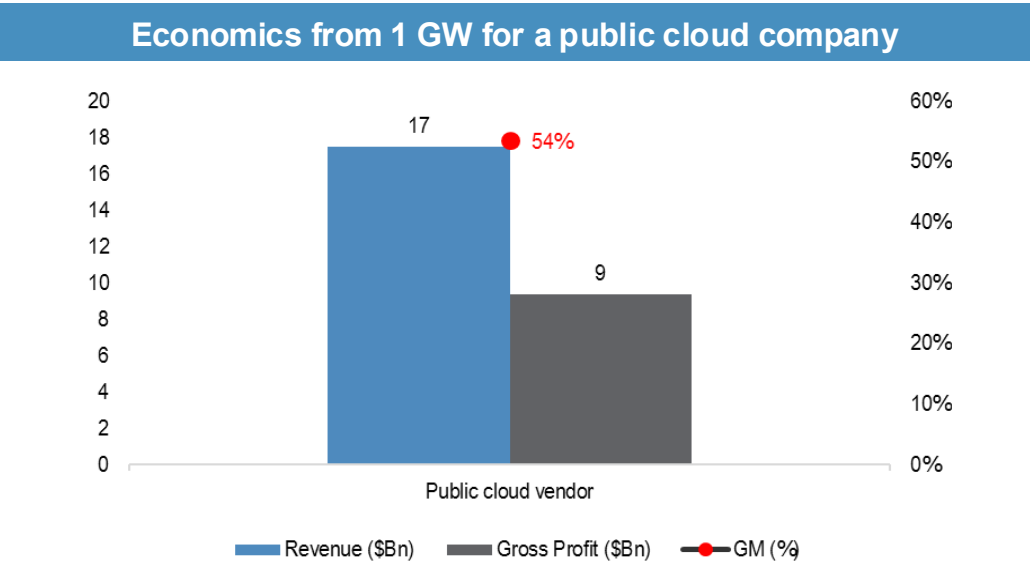
- Model/Token Seller with balanced token mix can earn~ \$31bn from a 1GW AI Token factory, up from \$10bn 1 year ago.
- Healthy Gross Margins given recent upswing in token pricing
- Public cloud vendor selling GPU compute can generate ~\$17bn per year from 1GW
- GPU rental pricing steadily moving up, with spot rates now well above existing contracts.
- Gross margins touching 50% for GPU rental models



Source: Bloomberg Finance L.P.



Source: J.P. Morgan estimates, Bloomberg.



Source: J.P. Morgan estimates, Bloomberg.

## Points of Contention in the Capital Markets regarding the AI Infrastructure Trade

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1. Open Source vs Proprietary Models – Who's Winning and how does it influence AI capex?
2. The Advent of RSI – Recursive Self-Improvement – Will it dampen or accelerate the AI infra investment cycle?
3. Sellers of Compute vs Buyers of Compute – How healthy are each of these camps?
4. Is there a Datacenter / Power Crunch, which will lead to an inventory of AI accelerators soon?
5. When can we see Wider adoption beyond Coding ?

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